

Steganography

CSC 3114 Cryptology and Coding Theory

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Lecture Outline

- 1 Meaning and purpose of steganography
- 2 Physical and digital techniques
- 3 Steganography compared with cryptography
- 4 Basic model and terminology
- 5 Text, image and audio steganography
- 6 Applications.

Guiding question

How can information be hidden in an ordinary carrier without creating noticeable evidence?

Learning Outcomes

By the end of this lecture, you should be able to:

- define steganography and explain its purpose;
- distinguish steganography from cryptography and watermarking;
- identify physical and digital information-hiding methods;
- describe the basic embedding and extraction model;
- explain text, image and audio steganography;
- discuss applications of steganography.

Steganography

Definition

Steganography is the art and science of hiding information inside an apparently innocent carrier so that an observer does not suspect that secret communication is taking place.

- The word comes from Greek: *steganos* (covered) and *graphein* (writing).
- A message may be hidden in an image, audio file, video, text, document or network communication.
- The visible carrier remains usable after the hidden data is inserted.

Assumption

Weak assumption

The system is secure because an attacker does not know which program or embedding method was used.

Stronger assumption

The attacker may know the algorithm. A secret stego-key controls the locations or rules used for embedding and extraction.

Lesson

Steganography should not depend only on “security through obscurity.”

Historical Examples

Historical examples include:

- messages written on wood and covered with wax;
- writing concealed on a messenger's body;
- invisible or secret inks;
- messages encoded in knitting or clothing patterns;
- writing placed beneath a postage stamp; and
- microdots hidden in ordinary correspondence.

Common principle

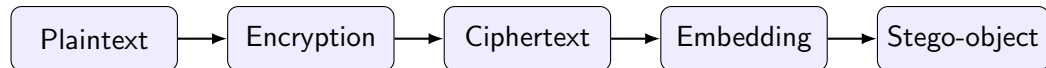
The hidden information occupies a location or representation that an observer is unlikely to inspect.

Cryptography vs. Steganography

Feature	Cryptography	Steganography
Primary goal	Conceal the meaning of a message	Conceal the existence of a message
Visible output	Ciphertext is recognizably encoded	Carrier should appear ordinary
Attacker's first task	Recover the plaintext or key	Determine whether hidden data exists
Output	Ciphertext	Stego-object
Failure	Plaintext is disclosed	Hidden channel is detected

Encryption and Steganography

Encryption and steganography work better together.



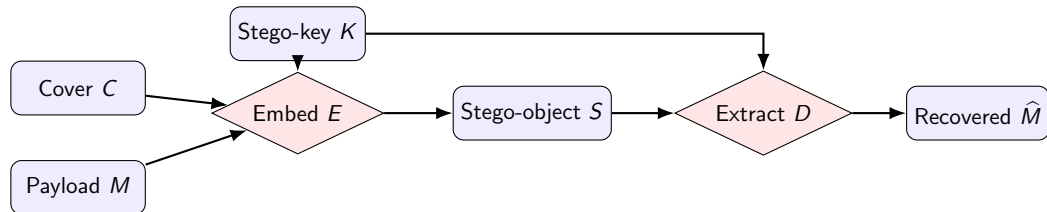
- Steganography makes communication less conspicuous.
- Encryption protects the message if the hidden data is discovered.
- Authentication can reveal whether the extracted payload was modified.

Watermarking vs. Steganography

	Steganography	Digital watermarking
Primary purpose	Covert communication	Ownership, authenticity or integrity
Relationship to carrier	Carrier transports the payload	Mark is meaningfully linked to the media
Desired survival	Depends on the communication channel	Often designed to resist editing or copying
Example	Secret text inside a photograph	Copyright identifier inside a video

Both belong to the broader field of **information hiding**.

The Basic Model: Embedding and Extraction Stages



$$S = E_K(C, M), \quad \hat{M} = D_K(S), \quad \hat{M} = M.$$

Key Terms in the Model

Term	Meaning
Cover object	Original carrier in which information will be hidden
Payload	Secret information to be concealed
Stego-key	Secret controlling embedding and extraction
Embedding algorithm	Procedure that inserts the payload into the cover
Stego-object	Carrier after the payload has been embedded
Extraction algorithm	Procedure that recovers the payload
Steganalysis	Detection, extraction or disruption of hidden information

Carrier Type Determines the Available Technique

Carrier	Common hiding feature	Major challenge
Text	Spacing, characters, syntax or word choice	Very little redundancy
Image	Pixel values or transform coefficients	Statistical detection and compression
Audio	Samples, phase, echo or frequency content	Human hearing and format conversion
Video	Frames, motion vectors or coefficients	Temporal consistency and transcoding
Network	Header fields, ordering or timing	Jitter, normalization and traffic analysis

An Acrostic Hides Data in Selected Characters

Read the first letter of each word:

Since **E**veryone **C**an **R**ead,
Encoding **T**ext **I**n **N**eutral
Sentences **I**s **D**oubtfully **E**ffective.

Hidden message

SECRET INSIDE

The carrier sentence is readable, but an unusual writing style may still create suspicion.

Text Encoding Needs an Explicit Rule

Suppose one space encodes 0 and two spaces encode 1.

Encoding 10110

Data__can__be__hidden__in__text.

The receiver must know:

- which spaces carry information;
- how bits are grouped into characters;
- where the payload begins and ends; and
- whether a stego-key changes the reading order.

Images Exploit Limits of Human Vision

An 8-bit RGB pixel contains three colour components:

$$P(x, y) = (R(x, y), G(x, y), B(x, y)), \quad R, G, B \in \{0, \dots, 255\}.$$

$R = 201$	$G = 118$	$B = 45$
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 $\implies$ one pixel

Small numerical changes are often difficult to see, especially in textured or noisy regions.

Image Capacity Can Be Estimated Before Embedding

For width W , height H , C colour channels and k embedded bits per channel:

$$\text{raw capacity} = W \times H \times C \times k \text{ bits.}$$

1024 × 768 RGB image using one LSB per channel

$$1024 \times 768 \times 3 = 2,359,296 \text{ bits} = 294,912 \text{ bytes} = 288 \text{ KiB.}$$

Headers, encryption metadata, authentication tags and error-correcting codes reduce the usable capacity.

Image Format Determines Whether the Payload Survives

Format	Opportunity	Main caution
BMP	Direct access to pixel bytes	Large uncompressed files can appear unusual
PNG	Lossless compression preserves pixels	Re-encoding may change colour representation
GIF	Palette indices can be modified	Palette changes may create visible errors
JPEG	Transform coefficients can carry bits	Direct pixel LSBs do not survive quantization

Audio Steganography

Audio payloads can be embedded in WAV, AU and compressed audio representations.

- **LSB coding:** modifies low-order bits of audio samples.
- **Echo hiding:** changes echo delay or amplitude slightly.
- **Phase coding:** alters phase relationships.
- **Spread spectrum:** distributes a weak signal across frequencies.

Design requirement

The method must match the expected channel: resampling, noise reduction and lossy compression may remove the payload.

Video and Networks Steganography

Video

Payloads may be placed in:

- individual frames;
- motion vectors;
- transform coefficients; or
- the audio track.

Networks

Covert channels may use:

- header fields;
- packet ordering;
- padding; or
- transmission timing.

Transcoding and network normalization can destroy these channels.

Applications of Steganography

Legitimate applications

- Confidential communication
- Media authentication
- Copyright protection
- Tamper detection
- Document provenance
- Database record linking

Potential misuse

- Concealing stolen information
- Hiding malware components
- Evading monitoring controls
- Covertly coordinating harmful activity

Steganalysis Searches for Evidence of Hidden Data

- **Structural analysis:** checks headers, metadata, file size and invalid fields.
- **Visual or auditory analysis:** looks for perceptible artifacts.
- **Statistical analysis:** tests distributions and correlations.
- **Machine learning:** classifies objects using features learned from examples.

Detection is a decision problem

A detector may produce false positives by labelling clean files as stego-objects, or false negatives by missing hidden data.

Questions?